Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Application software engineer with 2+ years building **Python services, REST integrations, and engineering data pipelines**; experienced translating customer challenges into custom solutions, visualizations, technical documentation, onboarding materials, and reliable cloud deployments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **Python REST services and data integrations** across FastAPI, PostgreSQL, SQL Server, and cloud storage, processing 5,000+ monthly workflows with validation, reconciliation, incremental loads, and 99%+ reliability.
- Translated user needs into actionable platform enhancements and dashboards, using indexed queries and analysis workflows to reduce latency from 1.2 seconds to 350 milliseconds while supporting 500+ users.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **autonomous-driving data pipelines** processing 100,000+ trajectory logs, combining HD-map geometry, temporal vehicle context, and simulation metrics to improve long-horizon planning 18% and closed-loop stability 15%.
- Analyzed experimental runs and visualized system behavior across 10,000 driving episodes, documenting constraints, anomalies, and performance findings for researchers to prioritize configuration changes and accelerate root-cause investigation.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Developed **custom API and data integrations** using Python, AWS Glue, S3, SQL, and caching, adapting multi-source workflows to partner requirements while improving response times 30% and efficiency 68%.
- Partnered with Microsoft and GitHub engineers to clarify integration challenges, troubleshoot failures, document findings, and deliver cross-platform builds with 20+ automated tests across Windows, Linux, and macOS.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **real-time data and visualization workflows** across eight Node.js services, WebSockets, REST APIs, and Azure SQL, synchronizing 2M+ records for resource, customer, billing, notification, and reporting tools.
- Investigated production behavior using Sentry, Azure Monitor, Kibana, and usage analytics, translating operational findings into tested enhancements that reduced resource wastage 42% and improved lead conversion 15%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Worked directly with customers to understand operational challenges, map 12+ integration points, and present technical designs for cross-platform customer booking, pricing, payment, analytics, and real-time status workflows.
- Delivered **reliable Python Flask APIs and data pipelines** with Firebase, secure webhooks, and real-time analytics, supporting 50,000+ records with sub-200-millisecond responses while improving cross-platform compatibility 15%.

Skills

Languages: Python, SQL, Java, JavaScript, TypeScript, C++, Bash
Integration: REST APIs, FastAPI, Flask, Node.js, WebSockets, webhooks, microservices, schema design
Data: PostgreSQL, SQL Server, Azure SQL, MongoDB, Firebase, ETL, streaming pipelines, data analysis
Visualization: React.js, dashboards, metrics, reporting, Google Analytics, Sentry, Azure Monitor, Kibana
Cloud & Delivery: AWS, GCP, Azure, Docker, Kubernetes, Git, Jenkins, CI/CD, technical documentation

Projects

CPU Scheduling Simulator Live | *C++, Metrics, Visualization*

- Implemented and analyzed **six C++ scheduling algorithms** with interactive controls, Gantt visualizations, edge-case handling, and live performance metrics across 1,000+ process combinations, reducing average turnaround time 25%.

FinNexus Live | *Real-Time Data, MongoDB, Kubernetes*

- Built a **real-time transaction data platform** supporting 50,000+ users and 1M+ transactions with secure third-party integrations, WebSocket updates, advanced filtering, MongoDB persistence, and Kubernetes-based production automation.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Enhanced a **cross-platform engineering data application** supporting 10+ SQL and NoSQL systems with schema exploration, structure synchronization, query generation, compatibility checks, Docker configurations, and reusable integration workflows.